

Experimental isobaric vapor–liquid equilibrium at atmospheric and sub-atmospheric pressures, excess molar volumes and deviations in molar refractivity from 293.15 K to 318.15 K of diisopropyl ether with methanol and isopropyl alcohol

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ABSTRACT

Experimental isobaric vapor–liquid equilibrium data for the binary systems, diisopropyl ether + isopropyl alcohol and methanol + diisopropyl ether at the local atmospheric pressure of 94.79 kPa and at sub-atmospheric pressures of (53.3, 66.7, 79.9) kPa were obtained over the entire composition range using a Sweitoslawsky-type ebulliometer. The experimental temperatures were compared with predictions made using Wilson, NRTL, UNIQUAC, UNIFAC, Peng–Robinson (PR) and Soave–Redlich–Kwong (SRK) equations of state (EoS) with Wong–Sandler (WS) mixing rules. Model parameters along with the deviations in temperature have been presented. Experimental temperatures were found to be in good agreement with the predicted values. NRTL model was found to represent the VLE behavior of these systems better than the other models used. Density and refractive indices of mixtures are reported from 293.15 K to 318.5 K.

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1. Introduction

The binary systems studied experimentally in this work, diisopropyl ether (DIPE) + isopropyl alcohol (IPA) and methanol + DIPE are part of the study of the vapor–liquid equilibrium (VLE) behavior of the ternary system, methanol + DIPE + IPA. DIPE is a potential eco-friendly gasoline additive [1] and an important solvent in the industry. It is found in the downstream of various industries such as in acetone manufacturing [2] and it can be synthesized from isopropanol. In the present study we have experimentally investigated the isobaric vapor–liquid equilibrium behavior of these binary systems at the local atmospheric pressure and sub-atmospheric pressures. Densities and refractive indices of the mixtures at various compositions have been measured from 293.15 K to 318.5 K. The work reported here is in continuation with our previous studies of phase equilibria and properties of industrially important streams containing alcohols, hydrocarbons and halogenated organics [3–5]. Chamorro et al. [1] have reported an azeotrope for the DIPE + IPA binary system as part

of the isothermal VLE data of the binary and ternary systems containing DIPE, IPA and *n*-heptane at 313.15 K and Lladosa et al. [6] have studied this system at 30 and 101.3 kPa. Frakova et al. [7] have reported isothermal VLE data for the methanol + DIPE system at 320 and 330 K.

2. Experimental

2.1. Materials Used

Methanol (>99.8 mass %, HPLC grade) provided by SD Fine Chemicals, India and diisopropyl ether (>98.5 mass %, GC grade) and isopropyl alcohol (>99.8 mass %, HPLC grade) supplied by Sigma–Aldrich were used for experimentation. The chemicals were stored in dessicators to prevent absorption of moisture and were used without any further purification. The final purity of the compounds was measured by a ZB-5 column using FID detector on a Shimadzu 2010 gas chromatograph. The densities and refractive indices of the pure components measured at 298.15 K are compared with literature values [8–12] in Table 1.

2.2. Apparatus

A Sweitoslawsky-type ebulliometer, similar to the one described by Hala et al. [13], was used for VLE measurements.

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Table 1Comparison of density, ρ and refractive index, n_D data with literature data at 94.79 kPa.

Chemical	IUPAC name	T (K)	ρ ($\times 10^3$ kg m $^{-3}$)		n_D	
			This work	Reference	This work	Reference
Methanol	Methanol	298.15	0.78662	0.78656 ^a 0.78637 ^b	1.32666	1.3267 ^a 1.32652 ^b
DIPE	2,2'-Oxybispropane	298.15	0.71830	0.71854 ^b 0.71823 ^c 0.71820 ^d 0.7183 ^e	1.3653	1.3655 ^c 1.3655 ^d 1.3652 ^e
IPA	2-propanol	298.15	0.78245	0.7827 ^e 0.78126 ^b 0.78130 ^f	1.37522	1.3754 ^e 1.3752 ^b 1.3750 ^f

DIPE: diisopropyl ether.

IPA: isopropyl alcohol.

 $u(\rho)$: ± 0.0001 kg m $^{-3}$. $u(n_D)$: ± 0.00001 .^a Ref. [8].^b Ref. [9].^c Ref. [10].^d Ref. [11].^e Ref. [6].^f Ref. [12].

The details of the construction of the ebulliometer and the experimental method used in the present study are as described by Vittal Prasad et al. [5]. The ebulliometer is connected to a vacuum pump and a nitrogen gas cylinder, along with a closed end manometer in line to maintain the pressure to within 0.05 kPa of the desired value. This is done by adjusting the opening of the needle valve of the gas cylinder or by opening the bypass line of the vacuum pump. A WIKA digital thermometer (Model CTH6200, ± 0.05 K) calibrated by point-to-point comparison with a WIKA hand-held calibrator (Model CEP3000) is used for measuring the equilibrium temperatures with an uncertainty of ± 0.05 K. The thermometer is placed in the thermowell containing glycerol to note the equilibrium temperature for the (vapor + liquid) mixture

impinging on the thermowell from the Cottrell tube of the ebulliometer. The equilibrium temperature is recorded after steady-state conditions, judged by the constancy of temperature and uniformity of the drop rate maintained for at least 1800s. The mixtures to be studied were prepared gravimetrically by weighing the required amounts of the pure liquids and stirring them well to make homogeneous mixtures before charging the mixtures into the still. A Mettler balance, capable of recording weights to the uncertainty of ± 0.0001 g, was used. After the mixture is charged in to the ebulliometer, vacuum is applied gradually and then the heating rate is slowly increased and adjusted to produce the required boil-up rate so that a drop count of about 30 drops per minute is achieved. This is in accordance with the suggestion of Hala et al. [13]. The VLE experiment is first done at the lowest pressure. After the measurement of the equilibrium temperature at this pressure, vacuum is released and the pressure increased to the

Table 2

Vapor pressures of IPA.

P_i^0 (kPa)	T (K)
14.02	312.63
16.09	315.22
18.94	318.35
21.65	320.97
24.56	323.46
26.99	325.36
29.69	327.30
32.53	329.19
36.08	331.35
38.28	332.60
40.49	333.80
43.63	335.42
45.88	336.52
48.77	337.86
51.47	339.06
54.24	340.23
56.82	341.27
59.59	342.35
62.14	343.30
65.69	344.58
67.60	345.24
70.27	346.13
73.16	347.08
75.65	347.87
78.35	348.70
81.05	349.51
83.80	350.31
86.40	351.05
96.28	353.69

 $u(T) = \pm 0.05$ K.**Table 3**

Vapor pressures of methanol.

P_i^0 (kPa)	T (K)
26.55	306.91
29.52	309.10
32.46	311.09
35.65	313.08
37.80	314.34
40.63	315.91
43.23	317.27
44.63	317.98
48.35	319.77
51.40	321.15
53.97	322.26
56.89	323.47
59.55	324.53
62.25	325.56
64.90	326.54
67.50	327.47
70.07	328.36
72.95	329.32
75.66	330.20
78.35	331.05
81.05	331.87
83.76	332.67
86.46	333.45
96.35	336.15

 $u(T) = \pm 0.05$ K.

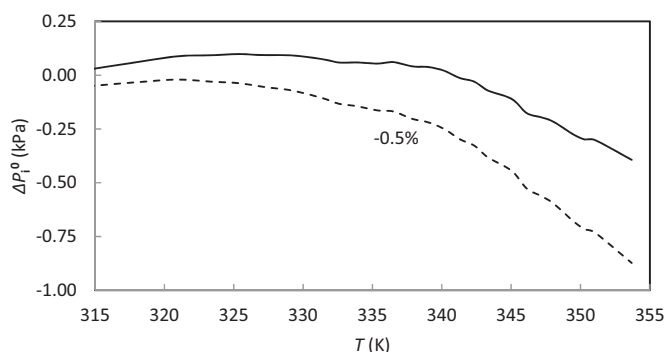


Fig. 1. $\Delta P_i^0 = P_{i,lit}^0 - P_{i,exp}^0$ in vapor pressures of IPA with Antoine constants from [14].

next higher value to be studied. Experimental VLE data were measured at (53.3, 66.7, 79.9 and 94.79) kPa.

The densities and refractive indices of the mixtures were measured with a Rudolph Research Analytical automatic densitometer (Model DDM 2911) and (Schmidt + Haensch) refractometer (Model DSR λ). The refractive index and density measurements of the binary mixtures were done from 293.15 to 318.15 K at intervals of 5 K. Refractive indices at (404.7, 435.8, 486.1, 546.1, 587.6, 589.3, 632.8, 644 and 706.5) nm wavelength frequencies along with the Abbe number were measured. The uncertainties of these instruments were determined to be 0.1 kg m^{-3} and 0.00001, respectively. The average deviation in mole fraction is ± 0.005 . The average uncertainty in the estimation of excess molar volumes is $\pm 1.14 \times 10^{-4} \text{ m}^3 \text{ kmol}^{-1}$. The standard uncertainty in measured pressure is $\pm 0.03 \text{ kPa}$.

3. Results and discussion

3.1. Pure component vapor pressures

Vapor pressures of the pure components IPA and methanol were measured in the same experimental set-up used for the VLE studies. The P – T data are given in Tables 2 and 3. The experimental pure component vapor pressures were found to be in good agreement with values obtained using Antoine constants given by Reid et al. [14] in Eq. (1).

$$\ln P_i^0 (\text{torr}) = A_i - \frac{B_i}{T(K) + C_i} \quad (1)$$

The deviation in the vapor pressures, $\Delta P_i^0 = P_{i,lit}^0 - P_{i,exp}^0$ is represented graphically in Figs. 1 and 2 for IPA and methanol respectively. It can be observed that the mean error is less than

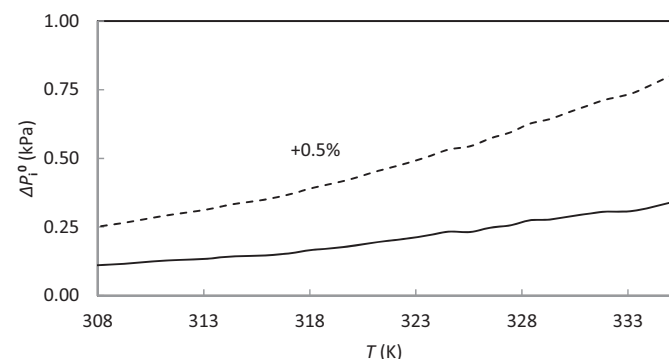


Fig. 2. $\Delta P_i^0 = P_{i,lit}^0 - P_{i,exp}^0$ in vapor pressures of methanol with Antoine constants from [14].

Table 4

Pure component data.

Chemical	T_c (K) ^a	P_c (kPa) ^a	ω^a	r^a	q^a	A^b	B^b	C^b
Methanol	512.6	80.9	0.556	1.4311	1.432	18.5875	3626.55	−34.29
DIPE	500.3	2880	0.331	5.4093	4.568	16.3417	2895.73	−43.15
IPA	508.3	4760	0.665	3.2491	3.124	18.6929	3640.2	−53.54

^a Ref. [15].

^b Ref. [14].

−0.5% for IPA and +0.5% for methanol, with respect to the literature values. The Antoine constants are listed in Table 4 along with other pure component data used in the data analysis.

3.2. Isobaric binary VLE

The experimental T – x_1 data of methanol + DIPE and DIPE + IPA binary systems at 53.3, 66.7, 79.9 and 94.79 kPa are reported in Tables 5 and 6. These were fitted to the Wilson, NRTL, UNIQUAC activity coefficient models, the group contribution method UNIFAC and Peng–Robinson (PR), Soave–Redlich–Kwong (SRK) Equations of State (EoS). The EoS were combined with the Wong–Sandler (WS) mixing rules. The data has also been correlated with the Legendre first-order polynomial. The pure component parameters such as critical temperature, critical pressure, acentric factor etc. and model equations for Wilson, NRTL, UNIQUAC and UNIFAC are

Table 5

Experimental VLE data of DIPE + IPA binary system.

x_1	T (K)	x_1	T (K)
$P = 53.33 \text{ kPa}$		$P = 66.67 \text{ kPa}$	
0	340.05	0	345.25
0.051	334.17	0.051	339.15
0.072	330.55	0.072	334.24
0.153	326.5	0.153	332.59
0.203	326.01	0.245	329.75
0.245	325.27	0.307	328.57
0.297	323.81	0.398	327.97
0.307	323.45	0.461	327.66
0.398	322.32	0.543	327.58
0.461	321.47	0.637	327.71
0.543	321.84	0.701	326.98
0.637	321.65	0.754	327.05
0.701	321.2	0.791	325.95
0.754	321.07	0.830	325.56
0.791	320.37	0.886	326.87
0.89	321	0.911	327.29
0.911	320.87	0.955	327.61
0.955	322.18	0.977	327.87
0.977	321.73	1.000	329.10
1	322.9	–	–
$P = 79.99 \text{ kPa}$		$P = 94.79 \text{ kPa}$	
0	349.5	0	354.5
0.051	341.61	0.051	345.39
0.153	335.32	0.072	343.55
0.194	335.17	0.153	340.38
0.3	334.1	0.194	339.23
0.307	333.32	0.300	338.04
0.398	333.24	0.398	338.38
0.461	332.49	0.461	338.17
0.543	332.56	0.543	337.40
0.637	332.64	0.637	337.63
0.701	332.22	0.701	337.10
0.754	331.65	0.754	336.90
0.791	331.03	0.791	335.19
0.83	330.57	0.830	335.50
0.886	332.15	0.886	336.46
0.911	332.41	0.911	336.83
0.955	332.32	0.955	337.05
1	334.3	1.000	340.40

$u(T) = \pm 0.05 \text{ K}$.

Table 6

Experimental VLE data of methanol + DIPE binary system.

x_1	$T(K)$	x_1	$T(K)$
$P = 53.33 \text{ kPa}$		$P = 66.67 \text{ kPa}$	
0	322.92	0	329.10
0.042	318.72	0.050	323.65
0.082	315.65	0.117	321.02
0.117	314.85	0.183	319.45
0.183	313.85	0.251	318.30
0.251	313.02	0.330	318.15
0.330	312.80	0.358	318.37
0.358	313.15	0.400	318.15
0.429	312.95	0.429	318.18
0.522	313.37	0.450	318.97
0.554	313.75	0.522	318.70
0.653	313.80	0.554	318.90
0.700	313.43	0.609	318.41
0.750	313.75	0.653	318.80
0.807	313.58	0.700	318.31
0.873	313.65	0.750	318.84
0.900	313.85	0.804	318.71
0.953	317.79	0.825	318.85
0.977	320.22	0.873	319.56
0.989	321.96	0.900	320.15
1	322.2	0.923	321.85
–	–	0.953	322.7
–	–	0.977	326.27
–	–	1	327.4
$P = 79.99 \text{ kPa}$		$P = 94.79 \text{ kPa}$	
0	334.35	0	339.4
0.117	325.13	0.117	330.64
0.183	323.95	0.183	328.60
0.251	323.12	0.251	329.59
0.330	323.09	0.330	327.63
0.358	322.83	0.400	328.65
0.400	323.33	0.429	327.38
0.429	323.00	0.450	327.98
0.450	323.72	0.522	327.64
0.522	322.97	0.609	327.73
0.566	323.72	0.653	327.75
0.609	323.39	0.750	328.22
0.653	323.68	0.837	327.70
0.700	323.55	0.873	328.29
0.750	324.09	0.900	328.27
0.837	323.82	0.925	331.06
0.873	324.14	0.989	335.42
0.900	323.79	1.000	335.98
0.953	325.39	–	–
1.000	331.8	–	–

 $u(T) = \pm 0.05 \text{ K}$

from Reid et al. [15]. The PR, SRK EoS and WS mixing rules equations are from Orbey and Sandler [16]. Legendre first-order polynomial with two parameters is based on the polynomial given by Gmehling and Onken [17].

Powell's Dog-Leg and Levenberg–Marquardt methods of non-linear regression as described by Madsen et al. [18] were used in the non-linear regression of experimental data to find optimum model parameters with the objective function,

$$\Phi = \left(\frac{P_{\text{cal}}}{P_{\text{cal}} - 1} \right)^2 \quad (2)$$

Powell's Dog-Leg method was found to be superior in all the calculations. Hence, the results from Dog-Leg method only have been presented here. The results of the correlation of experimental data with the models are presented in Tables 7 and 8 for DIPE + IPA and methanol + DIPE, respectively. The deviation of temperatures predicted by the different models from the experimentally measured values, expressed as root mean square deviation (rmsd),

represented by $\sigma(T)$ are presented along with the model parameters. Optimised parameters of the activity coefficient model NRTL have been used in the EoS model.

Among the different models used here, the DIPE + IPA binary system is best represented by the NRTL model, followed closely by the Wilson activity coefficient model and the PR EoS in combination with the WS mixing rules. For the methanol + DIPE binary system too, NRTL model gives the best fit. Both the systems exhibit minimum boiling azeotropes throughout the pressure range studied here. The effect of pressure on the VLE behavior of the two systems is represented graphically in Figs. 3 and 4.

The azeotropic conditions obtained from the NRTL model are presented in Table 9 along with literature data. The authors did not find literature data at the conditions studied in this work. Lladosa et al. [6] reported the VLE data of DIPE + IPA at 30 kPa and 101.3 kPa. The azeotropic temperature, pressure and compositions reported by Lladosa et al. [6], Verhoeve [19] and Yorizane et al. [20] are listed in Table 9. It can be observed that the azeotropic point shifts to the left with respect to the composition of the more volatile compound i.e. DIPE. Frakova et al. [7] reported isothermal VLE data of the methanol + DIPE binary system at 320 K and 330 K. The isobaric azeotropic data from this study calculated from NRTL model are reported in Table 9 along with the azeotropic data from Frakova et al. [7]. Both are in good agreement.

3.3. Excess molar volumes and deviations in molar refractivity

Densities and refractive indices of the mixtures over the entire composition range were measured from 293.15 to 318.15 K. Refractive indices were measured at (404.7, 435.8, 486.1, 546.1, 589.3, 632.8, 644 and 706.5) nm frequencies. Experimental densities and excess volumes of DIPE + IPA and methanol + DIPE mixtures at different compositions and temperatures are presented in Tables 10 and 11, respectively. The Abbe number values are also reported. The excess volumes and deviations in molar refractivity have been calculated from Eq. (3).

$$V^E(\text{m}^3\text{kmol}^{-1}) = \frac{\sum_i x_i M_i}{\rho_m} - \sum_i \frac{x_i M_i}{\rho_i} \quad (3)$$

The deviations in molar refractivity ΔR were calculated from Eq. (4) using the pure component ρ , n_D and molar refractivity (R_m).

$$\Delta R(\text{m}^3\text{kmol}^{-1}) = R_m - \sum_i \varphi_i R_i \quad (4)$$

$$R_m = \left(\frac{n_D^2 - 1}{n_D^2 + 1} \right) \left(\frac{\sum_i x_i M_i}{\rho_m} \right) \quad (5)$$

$$R_i = \left(\frac{n_{D,i}^2 - 1}{n_{D,i}^2 + 1} \right) \left(\frac{M_i}{\rho_i} \right) \quad (6)$$

$$\varphi_i = \frac{x_i V_i}{\sum_j x_j V_j} \quad (7)$$

where φ_i , n_D , $n_{D,i}$ and V_i are volume fraction of pure component i in the mixture, refractive index of mixture, refractive index of pure component and pure component molar volume, respectively. Refractive indices at different frequencies measured from 293.15 to 318.15 K are presented in Tables 12 and 13 for DIPE + IPA and methanol + IPA mixtures, respectively.

The four-parameter Redlich–Kister polynomial [21] was used to correlate the experimental V^E and ΔR data. The order of the

Table 7

Representation of VLE measurements by the different models for DIPE + IPA.

Parameter ^a	Wilson	NRTL	UNIQUAC	UNIFAC	PR + WS	SRK + WS	Legendre
<i>P</i> = 53.3 kPa							
<i>A</i> ₁₂	0.3668	1.0758	11.121	–	0.113	0.931	1.09
<i>A</i> ₂₁	0.3659	1.0737	133.335	–	–	–	1.09
α	–	0.597	–	–	–	–	–
σ (T) ^b	0.5494	0.4871	0.9713	11.0012	0.5503	1.6228	3.4859
<i>P</i> = 66.7 kPa							
<i>A</i> ₁₂	0.3172	1.0398	2.4199	–	0.132	0.963	1.04
<i>A</i> ₂₁	0.4109	1.2634	148.4757	–	–	–	1.04
α	–	0.6324	–	–	–	–	–
σ (T) ^b	0.7445	0.7164	1.2259	10.4166	0.7448	1.8428	3.5912
<i>P</i> = 79.9 kPa							
<i>A</i> ₁₂	0.2144	0.8283	–5.7233	–	0.14	0.945	1.04
<i>A</i> ₂₁	0.5338	1.4136	161.1647	–	–	–	1.04
α	–	0.5806	–	–	–	–	–
σ (T) ^b	0.6042	0.5558	0.9713	11.2029	0.5937	1.792	3.7598
<i>P</i> = 94.79 kPa							
<i>A</i> ₁₂	0.1996	1.0021	0.1187	–	0.16	0.946	0.9335
<i>A</i> ₂₁	0.5948	1.665	150.7639	–	–	–	0.9367
α	–	0.7097	–	–	–	–	–
σ (T) ^b	0.803	0.7343	1.0704	10.7434	0.7977	2.3501	4.3775

^a Wilson: $A_{12} = \Lambda_{12}$, $A_{21} = \Lambda_{21}$; NRTL: $A_{12} = \tau_{12}$, $A_{21} = \tau_{21}$; UNIQUAC: $A_{12} = (U_{12} - U_{22})/R$, $A_{21} = (U_{21} - U_{11})/R$; PR/SRK + WS: $A_{12} = k_{12} = k_{21}$; Legendre: $A_{12} = a_0$, $A_{21} = a_1$.^b σ (T) = $[\sum(T_{\text{exp}} - T_{\text{cal}})^2/n]^{0.5}$ where n is the number of observations.

polynomial has been determined by the *F*-test. The equation is as follows.

$$V_{12}^E \text{ or } \Delta R (\text{m}^3 \text{ kmol}^{-1}) = x_1 x_2 \sum_{i=1}^n A_i (x_1 - x_2)^{i-1} \quad (8)$$

The deviation in experimental and predicted values of V^E and ΔR represented in terms of standard deviation (SD) of the calculated values from the measured values. The SD values are given in Table 7 along with the Redlich–Kister parameters. The standard deviation is defined as

$$\text{SD} (\text{m}^3 \text{ kmol}^{-1}) = \left[\frac{\sum_i ((V^E \text{ or } \Delta R)_{\text{cal}} - (V^E \text{ or } \Delta R)_{\text{exp}})^2}{N - n} \right]^{0.5} \quad (9)$$

where N is the number of experimental data points and n is the number of fitted parameters. The correlation results for V^E and ΔR are relatively good as seen in Figs. 5–8. The Redlich–Kister parameters and the SD obtained for V^E and ΔR for both the both binary systems at the different temperatures studied are presented in Tables 14 and 15. It can be observed from Figs. 5–8 that while V^E

Table 8

Representation of VLE measurements by the different models for Methanol+DIPE.

Parameter ^a	Wilson	NRTL	UNIQUAC	UNIFAC	PR + WS	SRK + WS	Legendre
<i>P</i> = 53.3 kPa							
<i>A</i> ₁₂	0.3524	1.4663	18.0757	–	0.317	1.071	1.24
<i>A</i> ₂₁	0.2825	1.3868	367.6793	–	–	–	1.24
α	–	0.6357	–	–	–	–	–
σ (T) ^b	0.7731	0.7093	1.3457	9.2199	0.8134	3.1215	3.2788
<i>P</i> = 66.7 kPa							
<i>A</i> ₁₂	0.3615	1.3575	1.7481	–	0.319	1.06	1.259
<i>A</i> ₂₁	0.2939	1.2342	381.5954	–	–	–	1.2596
α	–	0.6016	–	–	–	–	–
σ (T) ^b	0.5774	0.5453	1.1902	10.4906	0.6829	2.6179	2.8479
<i>P</i> = 79.9 kPa							
<i>A</i> ₁₂	0.4844	2.6581	30.2693	–	0.31	1.031	1.17
<i>A</i> ₂₁	0.22	2.6738	336.5048	–	–	–	1.17
α	–	0.6679	–	–	–	–	–
σ (T) ^b	0.9091	0.4315	1.4059	9.9184	0.6451	3.1499	2.6711
<i>P</i> = 94.79 kPa							
<i>A</i> ₁₂	0.3924	1.7065	36.3901	–	0.31	1.016	1.2061
<i>A</i> ₂₁	0.3248	1.2752	317.3949	–	–	–	1.2084
α	–	0.713	–	–	–	–	–
σ (T) ^b	0.8646	0.7178	1.4236	9.2199	0.8849	2.5593	2.8014

^a Wilson : $A_{12} = \Lambda_{12}$, $A_{21} = \Lambda_{21}$; NRTL : $A_{12} = \tau_{12}$, $A_{21} = \tau_{21}$; UNIQUAC : $A_{12} = (U_{12} - U_{22})/R$, $A_{21} = (U_{21} - U_{11})/R$; PR/SRK + WS : $A_{12} = k_{12} = k_{21}$; Legendre : $A_{12} = a_0$, $A_{21} = a_1$.^b σ (T) = $[\sum(T_{\text{exp}} - T_{\text{cal}})^2/n]^{0.5}$ where n is the number of observations.

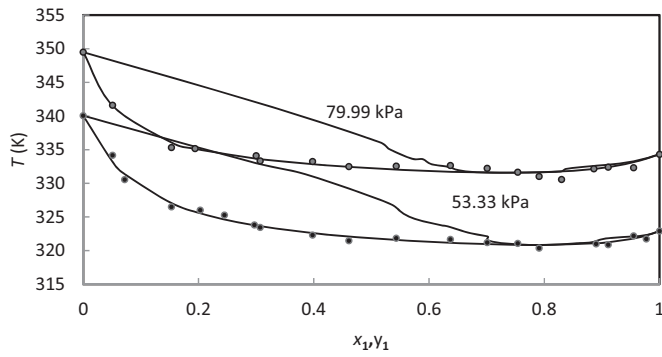


Fig. 3. Experimental data for DIPE+IPA system at 53.33 kPa and 79.99 kPa. (●) Experimental data; (—) calculated using NRTL model parameters given in Table 7.

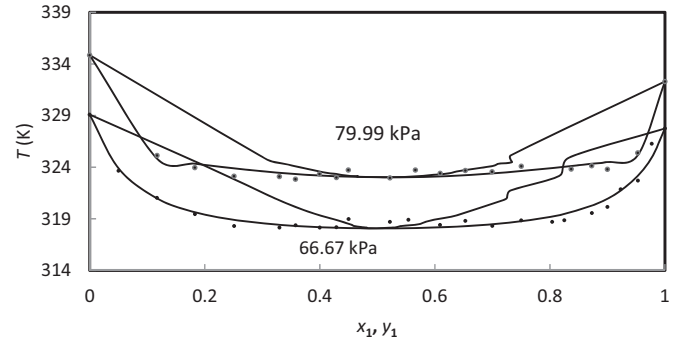


Fig. 4. Experimental data for Methanol+DIPE system at 66.67 kPa and 79.99 kPa. (●) Experimental data; (—) calculated using NRTL model parameters given in Table 8.

Table 9

Azeotropic compositions and temperatures.

System	P (kPa)	T (K)	x_1
DIPE + IPA	30.00	307.25 ^a	0.9064 ^a
	53.33	320.5 ^b	0.791 ^b
	66.67	325.56 ^b	0.767 ^b
	79.99	330.57 ^b	0.73 ^b
	98.49	335.5 ^b	0.73 ^b
	101.3	339.3 ^a	0.7818 ^a
	101.3	339.35 ^c	0.787 ^c
	101.3	339.25 ^d	0.8 ^d
Methanol + DIPE	53.33	313.15 ^b	0.476 ^b
	66.67	318.85 ^b	0.481 ^b
	68.47	320.00 ^e	0.5 ^e
	79.99	322.95 ^b	0.5225 ^b
	98.49	327.60 ^b	0.545 ^b
	101.61	330.00 ^e	0.539 ^e

^a Ref. [6].

^b This work.

^c Ref. [18].

^d Ref. [19].

^e Ref. [7].

Table 10

Densities, ρ ($\times 10^3$ kg m⁻³) and excess volumes, V^E ($\times 10^{-3}$ m³ kmol⁻¹) of DIPE+IPA mixtures at different temperatures at 94.79 kPa.

293.15 K			298.15 K			303.15 K			308.15 K			313.15 K			318.15 K		
x_1	ρ_m	V^E	ρ_m	V^E	ρ_m	V^E	ρ_m	V^E	ρ_m	V^E	ρ_m	V^E	ρ_m	V^E	ρ_m	V^E	V^E
0.0000	0.78691	0.00000	0.78245	0.00000	0.77784	0.00000	0.77313	0.00000	0.76847	0.00000	0.76379	0.00000	0.75923	-0.16050	0.75479	-0.16190	0.75039
0.0277	0.78454	-0.07760	0.78015	-0.09236	0.77528	-0.07240	0.77051	-0.07247	0.76620	-0.11489	0.76190	-0.11489	0.75761	-0.16190	0.75332	-0.16190	0.74903
0.0507	0.78314	-0.19205	0.77852	-0.18924	0.77370	-0.17990	0.76905	-0.19883	0.76454	-0.22723	0.75923	-0.22723	0.75479	-0.16050	0.75039	-0.16190	0.74600
0.0510	0.78312	-0.19362	—	—	0.77367	-0.18138	0.76905	-0.19883	0.76452	-0.22878	0.75921	-0.22878	0.75479	-0.16190	0.75039	-0.16190	0.74600
0.0717	0.78173	-0.27983	0.77695	-0.26597	0.77214	-0.26234	0.76756	-0.29387	0.76292	-0.31350	0.75756	-0.31350	0.75307	-0.23989	0.74817	-0.23989	0.74368
0.1528	0.77551	-0.48461	0.77053	-0.46665	0.76548	-0.45418	0.76084	-0.49712	0.75603	-0.51526	0.75075	-0.51526	0.74546	-0.44613	0.74016	-0.44613	0.73497
0.1938	0.77205	-0.52315	—	—	0.76188	-0.48974	0.75706	-0.51959	—	—	—	—	—	—	—	—	—
0.2029	0.77128	-0.52703	0.76641	-0.53035	0.76107	-0.49320	0.75620	-0.51875	0.75149	-0.55725	0.74641	-0.55725	0.74192	-0.50983	0.73693	-0.50983	0.73244
0.2450	0.76766	-0.52725	0.76295	-0.55794	0.75734	-0.49236	0.75220	-0.49239	0.74764	-0.55751	0.74279	-0.55751	0.73830	-0.53557	0.73381	-0.53557	0.72932
0.2970	—	—	0.75878	-0.56938	0.75277	-0.46274	0.74730	-0.42176	0.74294	-0.52802	0.73839	-0.52802	0.73390	-0.54269	0.72941	-0.54269	0.72492
0.3068	0.76242	-0.48993	0.75801	-0.56961	0.75196	-0.45515	0.74639	-0.40525	0.74211	-0.52036	0.73761	-0.52036	0.73312	-0.54189	0.72863	-0.54189	0.72414
0.3980	0.75524	-0.40649	0.75123	-0.55985	0.74462	-0.37244	0.73838	-0.23704	0.73456	-0.43631	0.73051	-0.43631	0.72602	-0.51969	0.72153	-0.51969	0.71704
0.4609	0.75087	-0.36456	0.74702	-0.55734	0.74013	-0.32953	0.73351	-0.13853	0.72995	-0.39281	0.72609	-0.39281	0.72160	-0.50646	0.71711	-0.50646	0.71262
0.5434	0.74599	-0.36240	0.74215	-0.58011	0.73509	-0.32289	0.72812	-0.07699	0.72476	-0.38711	0.72095	-0.38711	0.71646	-0.51393	0.71197	-0.51393	0.70748
0.6032	0.74303	-0.40658	0.73905	-0.61933	0.73200	-0.36170	0.72492	-0.09510	0.72158	-0.42811	0.71767	-0.42811	0.71318	-0.54200	0.70819	-0.54200	0.70370
0.6371	0.74154	-0.44747	0.73743	-0.64871	0.73043	-0.39908	0.72333	-0.12909	0.71996	-0.46723	0.71594	-0.46723	0.71145	-0.56523	0.70696	-0.56523	0.70247
0.7010	0.73899	-0.54511	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
0.7540	0.73702	-0.62869	0.73228	-0.75344	0.72568	-0.57075	0.71881	-0.33577	0.71506	-0.64559	0.71050	-0.64559	0.70591	-0.65079	0.70092	-0.65079	0.69643
0.7910	—	—	0.73068	-0.76969	0.72425	-0.61650	0.71751	-0.40537	0.71358	-0.69258	0.70882	-0.69258	0.70433	-0.66175	0.70003	-0.66175	0.69554
0.8296	0.73410	-0.69723	0.72895	-0.76445	0.72269	-0.64261	0.71611	-0.46052	0.71197	-0.71870	0.70702	-0.71870	0.70253	-0.65137	0.69754	-0.65137	0.69305
0.8903	0.73131	-0.64361	0.72597	-0.68308	0.71993	-0.60385	0.71361	-0.47019	0.70912	-0.67559	0.70397	-0.67559	0.69948	-0.56249	0.69499	-0.56249	0.69050
0.9113	0.73018	-0.58790	0.72483	-0.62581	0.71884	-0.55702	0.71260	-0.43911	0.70799	-0.62548	0.70281	-0.62548	0.69832	-0.50269	0.69383	-0.50269	0.68934
0.9547	0.72749	-0.38927	0.72220	-0.44392	0.71627	-0.38465	0.71018	-0.29374	0.70535	-0.44243	0.70018	-0.44243	0.69569	-0.31553	0.69070	-0.31553	0.68621

Table 10 (Continued)

293.15 K			298.15 K		303.15 K		308.15 K		313.15 K		318.15 K	
x_1	ρ_m	V^E	ρ_m	V^E	ρ_m	V^E	ρ_m	V^E	ρ_m	V^E	ρ_m	V^E
0.9730	0.72617	−0.26491	0.72098	−0.33678	0.71503	−0.27512	0.70899	−0.19213	0.70408	−0.32650	0.69896	−0.20608
0.9771	0.72586	−0.23286	0.72069	−0.30962	0.71473	−0.24679	0.70870	−0.16522	0.70378	−0.29653	0.69867	−0.17839
1.0000	0.72386	0.00000	0.71830	0.00000	0.71268	0.00000	0.70706	0.00000	0.70150	0.00000	0.69698	0.00000

$u(\rho) = \pm 0.1 \text{ kg m}^{-3}$.

Table 11

Densities, ρ ($\times 103 \text{ kg m}^{-3}$) and excess volumes, V^E ($\times 10^{-3} \text{ m}^3 \text{ kmol}^{-1}$) of methanol + DIPE mixtures at different temperatures at 94.79 kPa.

293.15			298.15		303.15		308.15		313.15		318.15	
x_1	ρ_m	V^E	ρ_m	V^E	ρ_m	V^E	ρ_m	V^E	ρ_m	V^E	ρ_m	V^E
0.0000	0.72386	0.00000	0.71830	0.00000	0.71268	0.00000	0.70706	0.00000	0.70150	0.00000	0.69698	0.00000
0.0423	0.72534	−0.11946	0.72012	−0.18570	–	–	0.70861	−0.13551	0.70292	−0.11035	0.69839	−0.11079
0.0424	0.72534	−0.11978	0.72013	−0.18599	0.71426	−0.14007	0.70863	−0.13924	0.70294	−0.11413	0.69841	−0.11463
0.0820	0.72736	−0.33042	0.72206	−0.38060	0.71626	−0.35158	0.71051	−0.32899	0.70495	−0.33262	0.69997	−0.24910
0.0822	0.72737	−0.33146	0.72207	−0.38160	0.71627	−0.35264	0.71052	−0.32995	0.70496	−0.33394	0.69998	−0.25013
0.1171	0.72904	−0.47828	0.72373	−0.52678	0.71796	−0.50509	0.71226	−0.49631	0.70679	−0.51757	0.70161	−0.40047
0.1271	0.72950	−0.51443	0.72419	−0.56391	0.71843	−0.54328	0.71274	−0.53720	0.70729	−0.56186	0.70207	−0.43901
0.1827	0.73199	−0.67841	0.72676	−0.73992	0.72098	−0.71920	0.71535	−0.72470	0.70993	−0.75493	0.70460	−0.62301
0.2162	0.73344	−0.75108	0.72828	−0.82169	0.72247	−0.79761	0.71686	−0.80665	0.71142	−0.83362	0.70610	−0.70996
0.2510	0.73494	−0.81068	0.72984	−0.88953	0.72399	−0.86068	0.71840	−0.87149	0.71293	−0.89223	0.70766	−0.78373
0.3298	0.73841	−0.90196	0.73339	−0.98734	0.72746	−0.94717	0.72188	−0.95863	0.71630	−0.96157	0.71124	−0.89766
0.3577	0.73969	−0.92359	–	–	0.72871	−0.96259	0.72314	−0.97398	0.71752	−0.97114	0.71255	−0.92289
0.4450	–	–	–	–	0.73304	−0.97210	–	–	–	–	–	–
0.4525	0.74438	−0.96864	0.73919	−1.01881	0.73318	−0.97156	0.72764	−0.98455	0.72196	−0.96951	0.71721	−0.96175
0.5220	0.74828	−0.97867	–	–	0.73682	−0.94614	0.73134	−0.96347	0.72566	−0.94808	0.72099	−0.95297
0.5543	0.75025	−0.97647	0.74460	−0.96577	0.73865	−0.92711	0.73322	−0.94747	0.72756	−0.93408	0.72288	−0.93986
0.6090	0.75385	−0.96074	0.74788	−0.91488	0.74201	−0.88529	0.73667	−0.91206	0.73107	−0.90408	0.72633	−0.90538
0.6529	0.75699	−0.93472	0.75075	−0.86370	0.74497	−0.84271	0.73973	−0.87547	0.73420	−0.87261	0.72937	−0.86634
0.7504	0.76478	−0.81901	0.75811	−0.71520	0.75258	−0.71362	0.74761	−0.76030	0.74224	−0.76541	0.73712	−0.73842
0.8040	0.76956	−0.71058	0.76286	−0.60868	0.75748	−0.61579	0.75269	−0.66904	–	–	–	–
0.8070	0.76984	−0.70338	0.76314	−0.60209	0.75778	−0.60939	–	–	0.74767	−0.66690	0.74240	−0.63127
0.8727	0.77623	−0.51127	0.76984	−0.43724	0.76466	−0.45020	0.76014	−0.50897	0.75477	−0.50310	0.74937	−0.46589
0.9253	0.78170	−0.30478	0.77596	−0.27176	0.77089	−0.28446	0.76658	−0.34421	0.76106	−0.32289	0.75570	−0.29320
0.9527	0.78468	−0.17596	0.77945	−0.17063	0.77442	−0.18125	0.77023	−0.24014	0.76457	−0.20797	0.75930	−0.18565
0.9773	0.78742	−0.04730	0.78277	−0.07000	0.77776	−0.07758	0.77367	−0.13481	0.76786	−0.09135	0.76270	−0.07763
0.9890	–	–	0.78441	−0.01870	0.77941	−0.02445	0.77536	−0.08058	0.76946	−0.03128	0.76438	−0.02227
0.9964	–	–	–	–	–	–	0.77646	−0.04453	0.77049	0.00864	–	–
1.0000	0.79162	0.00000	0.78662	0.00000	0.78154	0.00000	0.77650	0.00000	0.77153	0.00000	0.76660	0.00000

$u(\rho) = \pm 0.1 \text{ kg m}^{-3}$.

Table 12

Refractive indices, n_D of DIPE + IPA mixtures at different temperatures and different frequencies at 94.79 kPa.

293.5 K						n_D					
x_1	404.7	435.8	486.1	546.1	587.6	589.3	632.8	644	706.5	Abbe	
0	1.38830	1.38525	1.38179	1.37892	1.37742	1.37736	1.37605	1.37574	1.37424	59.10	
0.0277	1.38800	1.38493	1.38147	1.37860	1.37708	1.37703	1.37572	1.37541	1.37391	59.03	
0.0507	1.38790	1.38485	1.38137	1.37850	1.37699	1.37693	1.37562	1.37531	1.37381	58.95	
0.0510	1.38799	1.38492	1.38146	1.37859	1.37708	1.37702	1.37570	1.37539	1.37389	58.89	
0.0717	1.38774	1.38468	1.38121	1.37834	1.37683	1.37678	1.37545	1.37515	1.37366	58.95	
0.1528	1.38719	1.38413	1.38066	1.37779	1.37629	1.37623	1.37492	1.37461	1.37313	58.99	
0.1553	1.38719	1.38413	1.38066	1.37779	1.37629	1.37623	1.37492	1.37461	1.37313	58.92	
0.1938	1.38688	1.38381	1.38034	1.37747	1.37597	1.37591	1.37461	1.37430	1.37281	59.02	
0.2029	1.38669	1.38364	1.38016	1.37730	1.37580	1.37574	1.37443	1.37411	1.37263	58.83	
0.2450	1.38650	1.38325	1.37977	1.37692	1.37541	1.37536	1.37405	1.37374	1.37227	59.10	
0.2973	1.38627	1.38320	1.37973	1.37687	1.37535	1.37530	1.37400	1.37370	1.37222	59.10	
0.2970	1.38605	1.38300	1.37951	1.37666	1.37514	1.37509	1.37378	1.37347	1.37200	58.99	
0.3068	1.38571	1.38264	1.37919	1.37631	1.37481	1.37475	1.37345	1.37314	1.37167	58.82	
0.3980	1.38461	1.38154	1.37808	1.37522	1.37371	1.37366	1.37236	1.37206	1.37061	58.91	
0.4609	1.38404	1.38098	1.37753	1.37466	1.37315	1.37310	1.37180	1.37150	1.37005	58.80	
0.5434	1.38333	1.38026	1.37682	1.37395	1.37246	1.37240	1.37111	1.37078	1.36934	58.76	
0.6032	1.38277	1.37972	1.37624	1.37340	1.37188	1.37182	1.37052	1.37023	1.36879	58.70	
0.6371	1.38240	1.37934	1.37588	1.37304	1.37152	1.37146	1.37017	1.36988	1.36844	58.90	
0.7010	1.38179	1.37873	1.37528	1.37243	1.37093	1.37087	1.36958	1.36928	1.36784	59.04	
0.7540	1.38141	1.37834	1.37490	1.37205	1.37055	1.37049	1.36920	1.36891	1.36746	58.74	
0.7910	1.38107	1.37802	1.37455	1.37172	1.37022	1.37016	1.36887	1.36857	1.36713	58.79	

Table 12 (Continued)

293.5 K						n_D				
x_1	404.7	435.8	486.1	546.1	587.6	589.3	632.8	644	706.5	Abbe
0.8296	1.38071	1.37766	1.37423	1.37135	1.36987	1.36982	1.36854	1.36824	1.36680	58.57
0.8864	1.38051	1.37747	1.37400	1.37116	1.36966	1.36961	1.36832	1.36802	1.36658	58.71
0.8903	1.38011	1.37706	1.37363	1.37076	1.36928	1.36922	1.36795	1.36765	1.36620	58.56
0.9113	1.37989	1.37684	1.37339	1.37055	1.36905	1.36900	1.36771	1.36741	1.36598	58.66
0.9547	1.37948	1.37645	1.37301	1.37015	1.36866	1.36861	1.36732	1.36702	1.36559	58.37
0.9730	1.37918	1.37614	1.37271	1.36984	1.36836	1.36830	1.36703	1.36673	1.36529	58.39
0.9771	1.37921	1.37617	1.37274	1.36988	1.36839	1.36834	1.36706	1.36675	1.36532	58.35
1.0000	1.37888	1.37585	1.37241	1.36955	1.36806	1.36801	1.36674	1.36644	1.36501	58.54
298.15 K						n_D				
x_1	404.7	435.8	486.1	546.1	587.6	589.3	632.8	644	706.5	Abbe
0	1.38610	1.38307	1.37963	1.37678	1.37527	1.37522	1.37392	1.37361	1.37214	59.18
0.0277	1.38582	1.38277	1.37932	1.37648	1.37498	1.37492	1.37360	1.37329	1.37183	58.95
0.0507	1.38573	1.38268	1.37925	1.37639	1.37488	1.37483	1.37352	1.37322	1.37174	59.00
0.0510	1.38574	1.38276	1.37929	1.37644	1.37491	1.37489	1.37358	1.37327	1.37179	59.07
0.0717	1.38551	1.38247	1.37903	1.37618	1.37468	1.37462	1.37331	1.37300	1.37152	58.90
0.1528	1.38491	1.38186	1.37841	1.37557	1.37406	1.37401	1.37270	1.37240	1.37093	59.00
0.1553	1.38491	1.38186	1.37841	1.37557	1.37406	1.37401	1.37270	1.37240	1.37093	59.08
0.1938	1.38463	1.38158	1.37814	1.37530	1.37379	1.37373	1.37243	1.37213	1.37068	59.08
0.2029	1.38436	1.38132	1.37787	1.37502	1.37351	1.37346	1.37215	1.37185	1.37039	58.95
0.2450	1.38392	1.38088	1.37746	1.37462	1.37311	1.37306	1.37177	1.37147	1.37002	59.13
0.2973	1.38372	1.38067	1.37725	1.37439	1.37289	1.37284	1.37156	1.37126	1.36980	59.06
0.2970	1.38349	1.38044	1.37700	1.37426	1.37265	1.37259	1.37130	1.37100	1.36955	59.09
0.3068	1.38326	1.38023	1.37678	1.37394	1.37243	1.37238	1.37108	1.37078	1.36933	59.02
0.3980	1.38211	1.37907	1.37562	1.37278	1.37127	1.37122	1.36993	1.36964	1.36819	59.06
0.4609	1.38150	1.37845	1.37502	1.37217	1.37067	1.37063	1.36933	1.36903	1.36758	58.75
0.5434	1.38067	1.37764	1.37420	1.37135	1.36987	1.36981	1.36853	1.36823	1.36678	58.88
0.6032	1.38010	1.37707	1.37364	1.37080	1.36930	1.36925	1.36797	1.36767	1.36623	58.84
0.6371	1.37972	1.37669	1.37328	1.37042	1.36894	1.36889	1.36762	1.36731	1.36587	58.65
0.7010	1.37906	1.37604	1.37261	1.36978	1.36828	1.36823	1.36695	1.36666	1.36523	59.28
0.7540	1.37868	1.37567	1.37225	1.36940	1.36792	1.36786	1.36659	1.36629	1.36485	58.70
0.7910	1.37830	1.37529	1.37188	1.36903	1.36756	1.36750	1.36623	1.36593	1.36450	58.46
0.8296	1.37799	1.37498	1.37154	1.36872	1.36723	1.36717	1.36589	1.36559	1.36417	58.74
0.8864	1.37775	1.37471	1.37132	1.36847	1.36700	1.36695	1.36567	1.36537	1.36394	58.51
0.8903	1.37734	1.37434	1.37090	1.36807	1.36659	1.36653	1.36526	1.36497	1.36354	58.78
0.9113	1.37711	1.37411	1.37068	1.36785	1.36670	1.36631	1.36504	1.36475	1.36332	58.65
0.9547	1.37676	1.37367	1.37025	1.36742	1.36596	1.36587	1.36459	1.36430	1.36289	58.45
0.9730	1.37634	1.37334	1.36994	1.36710	1.36560	1.36557	1.36430	1.36400	1.36258	58.44
0.9771	1.37629	1.37330	1.36986	1.36704	1.36557	1.36551	1.36423	1.36393	1.36281	58.59
1.0000	1.37607	1.37308	1.36965	1.36683	1.36535	1.36530	1.36402	1.36373	1.36237	58.65
303.15 K						n_D				
x_1	404.7	435.8	486.1	546.1	587.6	589.3	632.8	644	706.5	Abbe
0	1.38379	1.38078	1.37738	1.37453	1.37304	1.37299	1.37170	1.37140	1.36993	59.12
0.0277	1.38362	1.38060	1.37718	1.37435	1.37285	1.37279	1.37149	1.37119	1.36973	59.07
0.0507	1.38359	1.38056	1.37714	1.37430	1.37280	1.37274	1.37145	1.37115	1.36969	59.01
0.0510	1.38351	1.38049	1.37707	1.37423	1.37273	1.37268	1.37138	1.37107	1.36962	58.97
0.0717	1.38322	1.38020	1.37678	1.37395	1.37244	1.37238	1.37109	1.37079	1.36934	59.16
0.1528	1.38258	1.37956	1.37614	1.37331	1.37181	1.37176	1.37047	1.37018	1.36872	59.20
0.1553	1.38258	1.37956	1.37614	1.37331	1.37181	1.37176	1.37047	1.37018	1.36872	59.16
0.1938	1.38229	1.37926	1.37584	1.37303	1.37153	1.37147	1.37018	1.36988	1.36843	59.18
0.2029	1.38193	1.37892	1.37549	1.37266	1.37116	1.37111	1.36981	1.36952	1.36807	59.06
0.2450	1.38148	1.37846	1.37506	1.37223	1.37075	1.37069	1.36941	1.36911	1.36767	59.18
0.2973	1.38111	1.37809	1.37467	1.37184	1.37036	1.37031	1.36902	1.36872	1.36727	59.05
0.2970	1.38091	1.37798	1.37446	1.37163	1.37013	1.37008	1.36880	1.36850	1.36705	58.99
0.3068	1.38076	1.37775	1.37433	1.37149	1.37001	1.36996	1.36867	1.36837	1.36692	58.99
0.3980	1.37951	1.37650	1.37309	1.37025	1.36877	1.36872	1.36744	1.36715	1.36570	58.90
0.4609	1.37885	1.37585	1.37245	1.36960	1.36818	1.36807	1.36680	1.36650	1.36501	58.74
0.5434	1.37802	1.37503	1.37161	1.36879	1.36731	1.36726	1.36598	1.36567	1.36425	58.76
0.6032	1.37738	1.37440	1.37096	1.36815	1.36667	1.36661	1.36533	1.36504	1.36362	58.83
0.6371	1.37701	1.37401	1.37059	1.36779	1.36631	1.36626	1.36497	1.36466	1.36326	58.78
0.7010	1.37631	1.37333	1.36992	1.36711	1.36564	1.36559	1.36431	1.36401	1.36259	59.43
0.7540	1.37593	1.37295	1.36954	1.36673	1.36525	1.36519	1.36391	1.36362	1.36221	58.66
0.7910	1.37553	1.37254	1.36913	1.36633	1.36485	1.36480	1.36352	1.36323	1.36182	58.73
0.8296	1.37518	1.37221	1.36880	1.36599	1.36453	1.36447	1.36320	1.36290	1.36148	58.65
0.8864	1.37495	1.37197	1.36856	1.36576	1.36430	1.36424	1.36297	1.36267	1.36126	58.65
0.8903	1.37451	1.37154	1.36813	1.36533	1.36386	1.36381	1.36254	1.36225	1.36083	58.74
0.9113	1.37428	1.37131	1.36791	1.36511	1.36365	1.36360	1.36233	1.36203	1.36061	58.65
0.9547	1.37382	1.37084	1.36747	1.36465	1.36319	1.36314	1.36187	1.36157	1.36016	58.43
0.9730	1.37347	1.37050	1.36711	1.36430	1.36285	1.36279	1.36153	1.36123	1.35983	58.56
0.9771	1.37338	1.37040	1.36702	1.36421	1.36275	1.36270	1.36145	1.36115	1.35972	58.61
1.0000	1.37321	1.37022	1.36687	1.36406	1.36260	1.36255	1.36130	1.36100	1.35960	58.57

308.15 K

x_1	404.7	435.8	486.1	546.1	587.6	589.3	632.8	644	706.5	Abbe
0	1.38152	1.37851	1.37512	1.37230	1.37080	1.37075	1.36946	1.36917	1.36772	59.25
0.0277	1.38135	1.37835	1.37495	1.37214	1.37065	1.37059	1.36929	1.36899	1.36755	59.04
0.0507	1.38130	1.37830	1.37488	1.37207	1.37059	1.37053	1.36924	1.36894	1.36748	59.15
0.0510	1.38124	1.37824	1.37483	1.37202	1.37053	1.37047	1.36917	1.36887	1.36742	58.94
0.0717	1.38092	1.37792	1.37451	1.37170	1.37072	1.37014	1.36886	1.36856	1.36711	59.21
0.1528	1.38023	1.37724	1.37384	1.37100	1.36953	1.36947	1.36820	1.36790	1.36646	58.92
0.1553	1.38023	1.37724	1.37384	1.37100	1.36953	1.36947	1.36820	1.36790	1.36646	59.09
0.1938	1.37994	1.37695	1.37356	1.37072	1.36925	1.36919	1.36792	1.36761	1.36617	58.95
0.2029	1.37930	1.37665	1.37311	1.37028	1.36880	1.36875	1.36748	1.36718	1.36573	58.94
0.2450	1.37901	1.37603	1.37266	1.36984	1.36837	1.36832	1.36706	1.36676	1.36532	59.15
0.2973	1.37852	1.37554	1.37215	1.36932	1.36785	1.36780	1.36653	1.36623	1.36480	58.93
0.2970	1.37832	1.37533	1.37193	1.36911	1.36764	1.36759	1.36631	1.36601	1.36457	58.85
0.3068	1.37823	1.37525	1.37180	1.36857	1.36754	1.36751	1.36624	1.36594	1.36451	58.92
0.3980	1.37692	1.37394	1.37056	1.36775	1.36628	1.36623	1.36496	1.36466	1.36323	58.89
0.4609	1.37625	1.37325	1.36987	1.36707	1.36560	1.36554	1.36427	1.36397	1.36254	58.79
0.5434	1.37535	1.37238	1.36896	1.36620	1.36473	1.36467	1.36338	1.36309	1.36168	58.98
0.6032	1.37466	1.37170	1.36830	1.36550	1.36404	1.36398	1.36271	1.36241	1.36100	58.68
0.6371	1.37427	1.37131	1.36792	1.36515	1.36369	1.36363	1.36235	1.36205	1.36064	58.86
0.7010	1.37352	1.37059	1.36722	1.36441	1.36297	1.36292	1.36166	1.36135	1.35993	59.39
0.7540	1.37312	1.37017	1.36681	1.36401	1.36256	1.36251	1.36125	1.36096	1.35954	58.82
0.7910	1.37271	1.36974	1.36638	1.36358	1.36215	1.36209	1.36085	1.36054	1.35911	58.70
0.8296	1.37234	1.36938	1.36604	1.36323	1.36178	1.36173	1.36048	1.36018	1.35877	58.56
0.8864	1.37212	1.36915	1.36580	1.36300	1.36156	1.36151	1.36026	1.35996	1.35855	58.58
0.8903	1.37167	1.36871	1.36535	1.36256	1.36111	1.36106	1.35981	1.35952	1.35810	58.84
0.9113	1.37142	1.36847	1.36511	1.36232	1.36088	1.36082	1.35957	1.35928	1.35786	58.65
0.9547	1.37095	1.36796	1.36465	1.36187	1.36041	1.36036	1.35910	1.35880	1.35741	58.49
0.9730	1.37062	1.36766	1.36431	1.36152	1.36008	1.36002	1.35877	1.35847	1.35707	58.38
0.9771	1.37048	1.36752	1.36417	1.36139	1.35994	1.35989	1.35864	1.35835	1.35694	58.61
1.0000	1.37035	1.36739	1.36407	1.36127	1.35975	1.35983	1.35853	1.35823	1.35683	58.42

313.15 K

x_1	404.7	435.8	486.1	546.1	587.6	589.3	632.8	644	706.5	Abbe
0	1.37921	1.37624	1.37288	1.37005	1.36857	1.36852	1.36726	1.36696	1.36551	59.10
0.0277	1.37904	1.37606	1.37269	1.36987	1.36840	1.36835	1.36708	1.36678	1.36533	59.09
0.0507	1.37898	1.37600	1.37262	1.36980	1.36833	1.36828	1.36700	1.36670	1.36524	58.99
0.0510	1.37891	1.37594	1.37256	1.36974	1.36828	1.36822	1.36696	1.36665	1.36520	59.10
0.0717	1.37855	1.37558	1.37221	1.36939	1.36792	1.36787	1.36659	1.36629	1.36485	58.89
0.1528	1.37787	1.37490	1.37150	1.36872	1.36724	1.36718	1.36590	1.36561	1.36418	59.22
0.1553	1.37787	1.37490	1.37150	1.36872	1.36724	1.36718	1.36590	1.36561	1.36418	59.09
0.1938	1.37761	1.37465	1.37126	1.36846	1.36699	1.36694	1.36566	1.36537	1.36394	59.13
0.2029	1.37706	1.37411	1.37068	1.36791	1.36645	1.36640	1.36512	1.36482	1.36337	59.28
0.2450	1.37653	1.37358	1.37022	1.36743	1.36596	1.36591	1.36465	1.36436	1.36294	59.25
0.2973	1.37596	1.37298	1.36959	1.36682	1.36535	1.36529	1.36401	1.36372	1.36230	59.14
0.2970	1.37571	1.37276	1.36938	1.36658	1.36512	1.36507	1.36381	1.36351	1.36209	59.03
0.3068	1.37567	1.37271	1.36934	1.36655	1.36509	1.36504	1.36377	1.36347	1.36204	58.97
0.3980	1.37428	1.37134	1.36798	1.36520	1.36376	1.36370	1.36244	1.36213	1.36071	58.93
0.4609	1.37355	1.37060	1.36724	1.36445	1.36302	1.36297	1.36172	1.36142	1.35999	59.03
0.5434	1.37263	1.36967	1.36633	1.36354	1.36211	1.36206	1.36081	1.36051	1.35908	58.85
0.6032	1.37187	1.36891	1.36561	1.36281	1.36136	1.36130	1.36007	1.35977	1.35834	58.76
0.6371	1.37152	1.36856	1.36522	1.36243	1.36099	1.36094	1.35970	1.35940	1.35800	58.84
0.7010	1.37074	1.36779	1.36447	1.36168	1.36025	1.36019	1.35895	1.35865	1.35723	59.40
0.7540	1.37030	1.36737	1.36403	1.36126	1.35981	1.35976	1.35851	1.35821	1.35681	58.75
0.7910	1.36988	1.36694	1.36363	1.36084	1.35941	1.35936	1.35812	1.35782	1.35641	58.61
0.8296	1.36948	1.36655	1.36322	1.36045	1.35902	1.35897	1.35772	1.35742	1.35602	58.59
0.8864	1.36937	1.36643	1.36308	1.36035	1.35890	1.35885	1.35758	1.35729	1.35589	58.72
0.8903	1.36880	1.36585	1.36252	1.35976	1.35833	1.35827	1.35704	1.35674	1.35534	58.76
0.9113	1.36853	1.36560	1.36228	1.35952	1.35809	1.35802	1.35679	1.35650	1.35510	58.92
0.9547	1.36805	1.36513	1.36180	1.35904	1.35761	1.35756	1.35631	1.35601	1.35460	58.54
0.9730	1.36771	1.36481	1.36145	1.35868	1.35727	1.35722	1.35599	1.35570	1.35426	58.78
0.9771	1.36756	1.36465	1.36134	1.35856	1.35715	1.35710	1.35587	1.35558	1.35416	58.62
1.0000	1.36740	1.36447	1.36119	1.35842	1.35700	1.35694	1.35572	1.35543	1.35401	58.66

318.15 K

x_1	404.7	435.8	486.1	546.1	587.6	589.3	632.8	644	706.5	Abbe
0	1.37691	1.37397	1.37061	1.36782	1.36635	1.36629	1.36502	1.36473	1.36329	59.14
0.0277	1.37668	1.37373	1.37036	1.36758	1.36611	1.36605	1.36479	1.36448	1.36305	59.07
0.0507	1.37662	1.37367	1.37030	1.36752	1.36604	1.36598	1.36472	1.36443	1.36300	59.22
0.0510	1.37654	1.37360	1.37024	1.36745	1.36598	1.36592	1.36464	1.36435	1.36292	58.99
0.0717	1.37616	1.37322	1.36986	1.36708	1.36563	1.36557	1.36428	1.36398	1.36256	59.05
0.1528	1.37550	1.37255	1.36919	1.36641	1.36495	1.36489	1.36364	1.36335	1.36190	59.24
0.1553	1.37550	1.37255	1.36919	1.36641	1.36495	1.36489	1.36364	1.36335	1.36190	59.19
0.1938	1.37525	1.37231	1.36894	1.36618	1.36472	1.36465	1.36338	1.36309	1.36166	59.22
0.2029	1.37457	1.37164	1.36826	1.36546	1.36403	1.36398	1.36273	1.36243	1.36100	59.19
0.2450	1.37400	1.37107	1.36775	1.36498	1.36354	1.36348	1.36223	1.36194	1.36053	59.33
0.2973	1.37334	1.37040	1.36706	1.36427	1.36283	1.36278	1.36153	1.36123	1.35982	58.93

Table 12 (Continued)

318.15 K										
x_1	404.7	435.8	486.1	546.1	587.6	589.3	632.8	644	706.5	Abbe
0.2970	1.37308	1.37014	1.36682	1.36402	1.36259	1.36254	1.36131	1.36101	1.35958	58.96
0.3068	1.37306	1.37013	1.36679	1.36402	1.36260	1.36255	1.36129	1.36098	1.35954	58.88
0.3980	1.37184	1.36891	1.36558	1.36283	1.36139	1.36134	1.36008	1.35978	1.35836	59.15
0.4609	1.37087	1.36796	1.36464	1.36184	1.36044	1.36036	1.35912	1.35883	1.35740	58.79
0.5434	1.36988	1.36693	1.36363	1.36086	1.35943	1.35938	1.35814	1.35784	1.35643	58.80
0.6032	1.36911	1.36618	1.36282	1.36011	1.35865	1.35860	1.35735	1.35707	1.35568	59.29
0.6371	1.36872	1.36578	1.36247	1.35971	1.35829	1.35824	1.35701	1.35671	1.35531	58.96
0.7010	1.36791	1.36501	1.36166	1.35892	1.35749	1.35743	1.35619	1.35591	1.35452	59.35
0.7540	1.36743	1.36453	1.36123	1.35847	1.35706	1.35700	1.35577	1.35547	1.35407	58.67
0.7910	1.36699	1.36408	1.36080	1.35802	1.35663	1.35658	1.35537	1.35507	1.35364	58.82
0.8296	1.36655	1.36364	1.36034	1.35760	1.35619	1.35614	1.35491	1.35461	1.35320	58.83
0.8864	1.36644	1.36355	1.36022	1.35745	1.35605	1.35599	1.35478	1.35448	1.35309	58.75
0.8903	1.36584	1.36294	1.35967	1.35689	1.35548	1.35543	1.35421	1.35391	1.35251	58.92
0.9113	1.36554	1.36264	1.35940	1.35663	1.35521	1.35515	1.35395	1.35366	1.35225	58.58
0.9547	1.36510	1.36222	1.35891	1.35618	1.35475	1.35469	1.35346	1.35317	1.35180	58.74
0.9730	1.36470	1.36182	1.35854	1.35580	1.35439	1.35433	1.35311	1.35281	1.35143	58.71
0.9771	1.36468	1.36178	1.35849	1.35575	1.35433	1.35427	1.35304	1.35276	1.35139	58.71
1.0000	1.36447	1.36157	1.35832	1.35556	1.35416	1.35411	1.35290	1.35259	1.35120	58.47

nm: nanometers.

 $u(n_D)$: ± 0.00001 .**Table 13**Refractive indices, n_D of DIPE + IPA mixtures at different temperatures and different frequencies at 94.79 kPa.

293.15 K					n_D					
x_1	404.7 nm	435.8 nm	486.1 nm	546.1 nm	587.6 nm	589.3 nm	632.8 nm	644 nm	706.5 nm	Abbe
0	1.38830	1.38525	1.38179	1.37892	1.37742	1.37736	1.37605	1.37574	1.37424	59.10
0.0277	1.38800	1.38493	1.38147	1.37860	1.37708	1.37703	1.37572	1.37541	1.37391	59.03
0.0507	1.38790	1.38485	1.38137	1.37850	1.37699	1.37693	1.37562	1.37531	1.37381	58.95
0.0510	1.38799	1.38492	1.38146	1.37859	1.37708	1.37702	1.37570	1.37539	1.37389	58.89
0.0717	1.38774	1.38468	1.38121	1.37834	1.37683	1.37678	1.37545	1.37515	1.37366	58.95
0.1528	1.38719	1.38413	1.38066	1.37779	1.37629	1.37623	1.37492	1.37461	1.37313	58.99
0.1553	1.38719	1.38413	1.38066	1.37779	1.37629	1.37623	1.37492	1.37461	1.37313	58.92
0.1938	1.38688	1.38381	1.38034	1.37747	1.37597	1.37591	1.37461	1.37430	1.37281	59.02
0.2029	1.38669	1.38364	1.38016	1.37730	1.37580	1.37574	1.37443	1.37411	1.37263	58.83
0.2450	1.38650	1.38325	1.37977	1.37692	1.37541	1.37536	1.37405	1.37374	1.37227	59.10
0.2973	1.38627	1.38320	1.37973	1.37687	1.37535	1.37530	1.37400	1.37370	1.37222	59.10
0.2970	1.38605	1.38300	1.37951	1.37666	1.37514	1.37509	1.37378	1.37347	1.37200	58.99
0.3068	1.38571	1.38264	1.37919	1.37631	1.37481	1.37475	1.37345	1.37314	1.37167	58.82
0.3980	1.38461	1.38154	1.37808	1.37522	1.37371	1.37366	1.37236	1.37206	1.37061	58.91
0.4609	1.38404	1.38098	1.37753	1.37466	1.37315	1.37310	1.37180	1.37150	1.37005	58.80
0.5434	1.38333	1.38026	1.37682	1.37395	1.37246	1.37240	1.37111	1.37078	1.36934	58.76
0.6032	1.38277	1.37972	1.37624	1.37340	1.37188	1.37182	1.37052	1.37023	1.36879	58.70
0.6371	1.38240	1.37934	1.37588	1.37304	1.37152	1.37146	1.37017	1.36988	1.36844	58.90
0.7010	1.38179	1.37873	1.37528	1.37243	1.37093	1.37087	1.36958	1.36928	1.36784	59.04
0.7540	1.38141	1.37834	1.37490	1.37205	1.37055	1.37049	1.36920	1.36891	1.36746	58.74
0.7910	1.38107	1.37802	1.37455	1.37172	1.37022	1.37016	1.36887	1.36857	1.36713	58.79
0.8296	1.38071	1.37766	1.37423	1.37135	1.36987	1.36982	1.36854	1.36824	1.36680	58.57
0.8864	1.38051	1.37747	1.37400	1.37116	1.36966	1.36961	1.36832	1.36802	1.36658	58.71
0.8903	1.38011	1.37706	1.37363	1.37076	1.36928	1.36922	1.36795	1.36765	1.36620	58.56
0.9113	1.37989	1.37684	1.37339	1.37055	1.36905	1.36900	1.36771	1.36741	1.36598	58.66
0.9547	1.37948	1.37645	1.37301	1.37015	1.36866	1.36861	1.36732	1.36702	1.36559	58.37
0.9730	1.37918	1.37614	1.37271	1.36984	1.36836	1.36830	1.36703	1.36673	1.36529	58.39
0.9771	1.37921	1.37617	1.37274	1.36988	1.36839	1.36834	1.36706	1.36675	1.36532	58.35
1.0000	1.37888	1.37585	1.37241	1.36955	1.36806	1.36801	1.36674	1.36644	1.36501	58.54

298.15 K

x_1	404.7 nm	435.8 nm	486.1 nm	546.1 nm	587.6 nm	589.3 nm	632.8 nm	644 nm	706.5 nm	Abbe
0	1.38610	1.38307	1.37963	1.37678	1.37527	1.37522	1.37392	1.37361	1.37214	59.18
0.0277	1.38582	1.38277	1.37932	1.37648	1.37498	1.37492	1.37360	1.37329	1.37183	58.95
0.0507	1.38573	1.38268	1.37925	1.37639	1.37488	1.37483	1.37352	1.37322	1.37174	59.00
0.0510	1.38574	1.38276	1.37929	1.37644	1.37491	1.37489	1.37358	1.37327	1.37179	59.07
0.0717	1.38551	1.38247	1.37903	1.37618	1.37468	1.37462	1.37331	1.37300	1.37152	58.90
0.1528	1.38491	1.38186	1.37841	1.37557	1.37406	1.37401	1.37270	1.37240	1.37093	59.00
0.1553	1.38491	1.38186	1.37841	1.37557	1.37406	1.37401	1.37270	1.37240	1.37093	59.08
0.1938	1.38463	1.38158	1.37814	1.37530	1.37379	1.37373	1.37243	1.37213	1.37068	59.08
0.2029	1.38436	1.38132	1.37787	1.37502	1.37351	1.37346	1.37215	1.37185	1.37039	58.95
0.2450	1.38392	1.38088	1.37746	1.37462	1.37311	1.37306	1.37177	1.37147	1.37002	59.13
0.2973	1.38372	1.38067	1.37725	1.37439	1.37289	1.37284	1.37156	1.37126	1.36980	59.06
0.2970	1.38349	1.38044	1.37700	1.37426	1.37265	1.37259	1.37130	1.37100	1.36955	59.09
0.3068	1.38326	1.38023	1.37678	1.37394	1.37243	1.37238	1.37108	1.37078	1.36933	59.02

Table 13 (Continued)

298.15 K										
x_1	404.7 nm	435.8 nm	486.1 nm	546.1 nm	587.6 nm	589.3 nm	632.8 nm	644 nm	706.5 nm	Abbe
0.3980	1.38211	1.37907	1.37562	1.37278	1.37127	1.37122	1.36993	1.36964	1.36819	59.06
0.4609	1.38150	1.37845	1.37502	1.37217	1.37067	1.37063	1.36933	1.36903	1.36758	58.75
0.5434	1.38067	1.37764	1.37420	1.37135	1.36987	1.36981	1.36853	1.36823	1.36678	58.88
0.6032	1.38010	1.37707	1.37364	1.37080	1.36930	1.36925	1.36797	1.36767	1.36623	58.84
0.6371	1.37972	1.37669	1.37328	1.37042	1.36894	1.36889	1.36762	1.36731	1.36587	58.65
0.7010	1.37906	1.37604	1.37261	1.36978	1.36828	1.36823	1.36695	1.36666	1.36523	59.28
0.7540	1.37868	1.37567	1.37225	1.36940	1.36792	1.36786	1.36659	1.36629	1.36485	58.70
0.7910	1.37830	1.37529	1.37188	1.36903	1.36756	1.36750	1.36623	1.36593	1.36450	58.46
0.8296	1.37799	1.37498	1.37154	1.36872	1.36723	1.36717	1.36589	1.36559	1.36417	58.74
0.8864	1.37775	1.37471	1.37132	1.36847	1.36700	1.36695	1.36567	1.36537	1.36394	58.51
0.8903	1.37734	1.37434	1.37090	1.36807	1.36659	1.36653	1.36526	1.36497	1.36354	58.78
0.9113	1.37711	1.37411	1.37068	1.36785	1.36670	1.36631	1.36504	1.36475	1.36332	58.65
0.9547	1.37676	1.37367	1.37025	1.36742	1.36596	1.36587	1.36459	1.36430	1.36289	58.45
0.9730	1.37634	1.37334	1.36994	1.36710	1.36560	1.36557	1.36430	1.36400	1.36258	58.44
0.9771	1.37629	1.37330	1.36986	1.36704	1.36557	1.36551	1.36423	1.36393	1.36281	58.59
1.0000	1.37607	1.37308	1.36965	1.36683	1.36535	1.36530	1.36402	1.36373	1.36237	58.65
303.15 K										
x_1	404.7 nm	435.8 nm	486.1 nm	546.1 nm	587.6 nm	589.3 nm	632.8 nm	644 nm	706.5 nm	Abbe
0	1.38379	1.38078	1.37738	1.37453	1.37304	1.37299	1.37170	1.37140	1.36993	59.12
0.0277	1.38362	1.38060	1.37718	1.37435	1.37285	1.37279	1.37149	1.37119	1.36973	59.07
0.0507	1.38359	1.38056	1.37714	1.37430	1.37280	1.37274	1.37145	1.37115	1.36969	59.01
0.0510	1.38351	1.38049	1.37707	1.37423	1.37273	1.37268	1.37138	1.37107	1.36962	58.97
0.0717	1.38322	1.38020	1.37678	1.37395	1.37244	1.37238	1.37109	1.37079	1.36934	59.16
0.1528	1.38258	1.37956	1.37614	1.37331	1.37181	1.37176	1.37047	1.37018	1.36872	59.20
0.1553	1.38258	1.37956	1.37614	1.37331	1.37181	1.37176	1.37047	1.37018	1.36872	59.16
0.1938	1.38229	1.37926	1.37584	1.37303	1.37153	1.37147	1.37018	1.36988	1.36843	59.18
0.2029	1.38193	1.37892	1.37549	1.37266	1.37116	1.37111	1.36981	1.36952	1.36807	59.06
0.2450	1.38148	1.37846	1.37506	1.37223	1.37075	1.37069	1.36941	1.36911	1.36767	59.18
0.2973	1.38111	1.37809	1.37467	1.37184	1.37036	1.37031	1.36902	1.36872	1.36727	59.05
0.2970	1.38091	1.37798	1.37446	1.37163	1.37013	1.37008	1.36880	1.36850	1.36705	58.99
0.3068	1.38076	1.37775	1.37433	1.37149	1.37001	1.36996	1.36867	1.36837	1.36692	58.99
0.3980	1.37951	1.37650	1.37309	1.37025	1.36877	1.36872	1.36744	1.36715	1.36570	58.90
0.4609	1.37885	1.37585	1.37245	1.36960	1.36818	1.36807	1.36680	1.36650	1.36501	58.74
0.5434	1.37802	1.37503	1.37161	1.36879	1.36731	1.36726	1.36598	1.36567	1.36425	58.76
0.6032	1.37738	1.37440	1.37096	1.36815	1.36667	1.36661	1.36533	1.36504	1.36362	58.83
0.6371	1.37701	1.37401	1.37059	1.36779	1.36631	1.36626	1.36497	1.36466	1.36326	58.78
0.7010	1.37631	1.37333	1.36992	1.36711	1.36564	1.36559	1.36431	1.36401	1.36259	59.43
0.7540	1.37593	1.37295	1.36954	1.36673	1.36525	1.36519	1.36391	1.36362	1.36221	58.66
0.7910	1.37553	1.37254	1.36913	1.36633	1.36485	1.36480	1.36352	1.36323	1.36182	58.73
0.8296	1.37518	1.37221	1.36880	1.36599	1.36453	1.36447	1.36320	1.36290	1.36148	58.65
0.8864	1.37495	1.37197	1.36856	1.36576	1.36430	1.36424	1.36297	1.36267	1.36126	58.65
0.8903	1.37451	1.37154	1.36813	1.36533	1.36386	1.36381	1.36254	1.36225	1.36083	58.74
0.9113	1.37428	1.37131	1.36791	1.36511	1.36365	1.36360	1.36233	1.36203	1.36061	58.65
0.9547	1.37382	1.37084	1.36747	1.36465	1.36319	1.36314	1.36187	1.36157	1.36016	58.43
0.9730	1.37347	1.37050	1.36711	1.36430	1.36285	1.36279	1.36153	1.36123	1.35983	58.56
0.9771	1.37338	1.37040	1.36702	1.36421	1.36275	1.36270	1.36145	1.36115	1.35972	58.61
1.0000	1.37321	1.37022	1.36687	1.36406	1.36260	1.36255	1.36130	1.36100	1.35960	58.57
308.15 K										
x_1	404.7 nm	435.8 nm	486.1 nm	546.1 nm	587.6 nm	589.3 nm	632.8 nm	644 nm	706.5 nm	Abbe
0	1.38152	1.37851	1.37512	1.37230	1.37080	1.37075	1.36946	1.36917	1.36772	59.25
0.0277	1.38135	1.37835	1.37495	1.37214	1.37065	1.37059	1.36929	1.36899	1.36755	59.04
0.0507	1.38130	1.37830	1.37488	1.37207	1.37059	1.37053	1.36924	1.36894	1.36748	59.15
0.0510	1.38124	1.37824	1.37483	1.37202	1.37053	1.37047	1.36917	1.36887	1.36742	58.94
0.0717	1.38092	1.37792	1.37451	1.37170	1.37022	1.37014	1.36886	1.36856	1.36711	59.21
0.1528	1.38023	1.37724	1.37384	1.37100	1.36953	1.36947	1.36820	1.36790	1.36646	58.92
0.1553	1.38023	1.37724	1.37384	1.37100	1.36953	1.36947	1.36820	1.36790	1.36646	59.09
0.1938	1.37994	1.37695	1.37356	1.37072	1.36925	1.36919	1.36792	1.36761	1.36617	58.95
0.2029	1.37930	1.37665	1.37311	1.37028	1.36880	1.36875	1.36748	1.36718	1.36573	58.94
0.2450	1.37901	1.37603	1.37266	1.36984	1.36837	1.36832	1.36706	1.36676	1.36532	59.15
0.2973	1.37852	1.37554	1.37215	1.36932	1.36785	1.36780	1.36653	1.36623	1.36480	58.93
0.2970	1.37832	1.37533	1.37193	1.36911	1.36764	1.36759	1.36631	1.36601	1.36457	58.85
0.3068	1.37823	1.37525	1.37180	1.36897	1.36754	1.36751	1.36624	1.36594	1.36451	58.92
0.3980	1.37692	1.37394	1.37056	1.36775	1.36628	1.36623	1.36496	1.36466	1.36323	58.89
0.4609	1.37625	1.37325	1.36987	1.36707	1.36560	1.36554	1.36427	1.36397	1.36254	58.79
0.5434	1.37535	1.37238	1.36896	1.36620	1.36473	1.36467	1.36338	1.36309	1.36168	58.98
0.6032	1.37466	1.37170	1.36830	1.36550	1.36404	1.36398	1.36271	1.36241	1.36100	58.68
0.6371	1.37427	1.37131	1.36792	1.36515	1.36369	1.36363	1.36235	1.36205	1.36064	58.86
0.7010	1.37352	1.37059	1.36722	1.36441	1.36297	1.36292	1.36166	1.36135	1.35993	59.39
0.7540	1.37312	1.37017	1.36681	1.36401	1.36256	1.36251	1.36125	1.36096	1.35954	58.82
0.7910	1.37271	1.36974	1.36638	1.36358	1.36215	1.36209	1.36085	1.36054	1.35911	58.70
0.8296	1.37234	1.36938	1.36604	1.36323	1.36178	1.36173	1.36048	1.36018	1.35877	58.56

Table 13 (Continued)

308.15 K										
x_1	404.7 nm	435.8 nm	486.1 nm	546.1 nm	587.6 nm	589.3 nm	632.8 nm	644 nm	706.5 nm	Abbe
0.8864	1.37212	1.36915	1.36580	1.36300	1.36156	1.36151	1.36026	1.35996	1.35855	58.58
0.8903	1.37167	1.36871	1.36535	1.36256	1.36111	1.36106	1.35981	1.35952	1.35810	58.84
0.9113	1.37142	1.36847	1.36511	1.36232	1.36088	1.36082	1.35957	1.35928	1.35786	58.65
0.9547	1.37095	1.36796	1.36465	1.36187	1.36041	1.36036	1.35910	1.35880	1.35741	58.49
0.9730	1.37062	1.36766	1.36431	1.36152	1.36008	1.36002	1.35877	1.35847	1.35707	58.38
0.9771	1.37048	1.36752	1.36417	1.36139	1.35994	1.35989	1.35864	1.35835	1.35694	58.61
1.0000	1.37035	1.36739	1.36407	1.36127	1.35975	1.35983	1.35853	1.35823	1.35683	58.42
313.15 K										
x_1	404.7 nm	435.8 nm	486.1 nm	546.1 nm	587.6 nm	589.3 nm	632.8 nm	644 nm	706.5 nm	Abbe
0	1.37921	1.37624	1.37288	1.37005	1.36857	1.36852	1.36726	1.36696	1.36551	59.10
0.0277	1.37904	1.37606	1.37269	1.36987	1.36840	1.36835	1.36708	1.36678	1.36533	59.09
0.0507	1.37898	1.37600	1.37262	1.36980	1.36833	1.36828	1.36700	1.36670	1.36524	58.99
0.0510	1.37891	1.37594	1.37256	1.36974	1.36828	1.36822	1.36696	1.36665	1.36520	59.10
0.0717	1.37855	1.37558	1.37221	1.36939	1.36792	1.36787	1.36659	1.36629	1.36485	58.89
0.1528	1.37787	1.37490	1.37150	1.36872	1.36724	1.36718	1.36590	1.36561	1.36418	59.22
0.1553	1.37787	1.37490	1.37150	1.36872	1.36724	1.36718	1.36590	1.36561	1.36418	59.09
0.1938	1.37761	1.37465	1.37126	1.36846	1.36699	1.36694	1.36566	1.36537	1.36394	59.13
0.2029	1.37706	1.37411	1.37068	1.36791	1.36645	1.36640	1.36512	1.36482	1.36337	59.28
0.2450	1.37653	1.37358	1.37022	1.36743	1.36596	1.36591	1.36465	1.36436	1.36294	59.25
0.2973	1.37596	1.37298	1.36959	1.36682	1.36535	1.36529	1.36401	1.36372	1.36230	59.14
0.2970	1.37571	1.37276	1.36938	1.36658	1.36512	1.36507	1.36381	1.36351	1.36209	59.03
0.3068	1.37567	1.37271	1.36934	1.36655	1.36509	1.36504	1.36377	1.36347	1.36204	58.97
0.3980	1.37428	1.37134	1.36798	1.36520	1.36376	1.36370	1.36244	1.36213	1.36071	58.93
0.4609	1.37355	1.37060	1.36724	1.36445	1.36302	1.36297	1.36172	1.36142	1.35999	59.03
0.5434	1.37263	1.36967	1.36633	1.36354	1.36211	1.36206	1.36081	1.36051	1.35908	58.85
0.6032	1.37187	1.36891	1.36561	1.36281	1.36136	1.36130	1.36007	1.35977	1.35834	58.76
0.6371	1.37152	1.36856	1.36522	1.36243	1.36099	1.36094	1.35970	1.35940	1.35800	58.84
0.7010	1.37074	1.36779	1.36447	1.36168	1.36025	1.36019	1.35895	1.35865	1.35723	59.40
0.7540	1.37030	1.36737	1.36403	1.36126	1.35981	1.35976	1.35851	1.35821	1.35681	58.75
0.7910	1.36988	1.36694	1.36363	1.36084	1.35941	1.35936	1.35812	1.35782	1.35641	58.61
0.8296	1.36948	1.36655	1.36322	1.36045	1.35902	1.35897	1.35772	1.35742	1.35602	58.59
0.8864	1.36937	1.36643	1.36308	1.36035	1.35890	1.35885	1.35758	1.35729	1.35589	58.72
0.8903	1.36880	1.36585	1.36252	1.35976	1.35833	1.35827	1.35704	1.35674	1.35534	58.76
0.9113	1.36853	1.36560	1.36228	1.35952	1.35809	1.35802	1.35679	1.35650	1.35510	58.92
0.9547	1.36805	1.36513	1.36180	1.35904	1.35761	1.35756	1.35631	1.35601	1.35460	58.54
0.9730	1.36771	1.36481	1.36145	1.35868	1.35727	1.35722	1.35599	1.35570	1.35426	58.78
0.9771	1.36756	1.36465	1.36134	1.35856	1.35715	1.35710	1.35587	1.35557	1.35416	58.62
1.0000	1.36740	1.36447	1.36119	1.35842	1.35700	1.35694	1.35572	1.35543	1.35401	58.66
318.15 K										
x_1	404.7 nm	435.8 nm	486.1 nm	546.1 nm	587.6 nm	589.3 nm	632.8 nm	644 nm	706.5 nm	Abbe
0	1.37691	1.37397	1.37061	1.36782	1.36635	1.36629	1.36502	1.36473	1.36329	59.14
0.0277	1.37668	1.37373	1.37036	1.36758	1.36611	1.36605	1.36479	1.36448	1.36305	59.07
0.0507	1.37662	1.37367	1.37030	1.36752	1.36604	1.36598	1.36472	1.36443	1.36300	59.22
0.0510	1.37654	1.37360	1.37024	1.36745	1.36598	1.36592	1.36464	1.36435	1.36292	58.99
0.0717	1.37616	1.37322	1.36986	1.36708	1.36563	1.36557	1.36428	1.36398	1.36256	59.05
0.1528	1.37550	1.37255	1.36919	1.36641	1.36495	1.36489	1.36364	1.36335	1.36190	59.24
0.1553	1.37550	1.37255	1.36919	1.36641	1.36495	1.36489	1.36364	1.36335	1.36190	59.19
0.1938	1.37525	1.37231	1.36894	1.36618	1.36470	1.36465	1.36338	1.36309	1.36166	59.22
0.2029	1.37457	1.37164	1.36826	1.36546	1.36403	1.36398	1.36273	1.36243	1.36100	59.19
0.2450	1.37400	1.37107	1.36775	1.36498	1.36354	1.36348	1.36223	1.36194	1.36053	59.33
0.2973	1.37334	1.37040	1.36706	1.36427	1.36283	1.36278	1.36153	1.36123	1.35982	58.93
0.2970	1.37308	1.37014	1.36682	1.36402	1.36259	1.36254	1.36131	1.36101	1.35958	58.96
0.3068	1.37306	1.37013	1.36679	1.36402	1.36260	1.36255	1.36129	1.36098	1.35954	58.88
0.3980	1.37184	1.36891	1.36558	1.36283	1.36139	1.36134	1.36008	1.35978	1.35836	59.15
0.4609	1.37087	1.36796	1.36464	1.36184	1.36044	1.36036	1.35912	1.35883	1.35740	58.79
0.5434	1.36988	1.36693	1.36363	1.36086	1.35943	1.35938	1.35814	1.35784	1.35643	58.80
0.6032	1.36911	1.36618	1.36282	1.36011	1.35865	1.35860	1.35735	1.35707	1.35568	59.29
0.6371	1.36872	1.36578	1.36247	1.35971	1.35829	1.35824	1.35701	1.35671	1.35531	58.96
0.7010	1.36791	1.36501	1.36166	1.35892	1.35749	1.35743	1.35619	1.35591	1.35452	59.35
0.7540	1.36743	1.36453	1.36123	1.35847	1.35706	1.35700	1.35577	1.35547	1.35407	58.67
0.7910	1.36699	1.36408	1.36080	1.35802	1.35663	1.35658	1.35537	1.35507	1.35364	58.82
0.8296	1.36655	1.36364	1.36034	1.35760	1.35619	1.35614	1.35491	1.35461	1.35320	58.83
0.8864	1.36644	1.36355	1.36022	1.35745	1.35605	1.35599	1.35478	1.35448	1.35309	58.75
0.8903	1.36584	1.36294	1.35967	1.35689	1.35548	1.35543	1.35421	1.35391	1.35251	58.92
0.9113	1.36554	1.36264	1.35940	1.35663	1.35521	1.35515	1.35395	1.35366	1.35225	58.58
0.9547	1.36510	1.36222	1.35891	1.35618	1.35475	1.35469	1.35346	1.35317	1.35180	58.74
0.9730	1.36470	1.36182	1.35854	1.35580	1.35439	1.35433	1.35311	1.35281	1.35143	58.71
0.9771	1.36468	1.36178	1.35849	1.35575	1.35433	1.35427	1.35304	1.35276	1.35139	58.71
1.0000	1.36447	1.36157	1.35832	1.35556	1.35416	1.35411	1.35290	1.35259	1.35120	58.47

nm = nanometers, $u(n_D) = \pm 0.00001$.

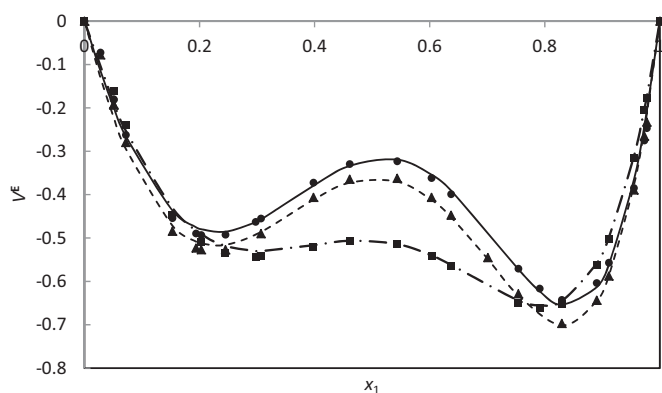


Fig. 5. V^E ($\times 10^{-3} \text{ m}^3 \text{ kmol}^{-1}$) for the binary system DIPE + IPA. ($\blacktriangle$) Experimental at 293.15 K, ($\bullet$) experimental data at 303.15 K, ($\blacksquare$) experimental at 318.15 K; (—) from Redlich–Kister parameters at 293.15 K, (—) solid curve from Redlich–Kister parameters 303.15–K, (---) from Redlich–Kister parameters at 318.15 K.

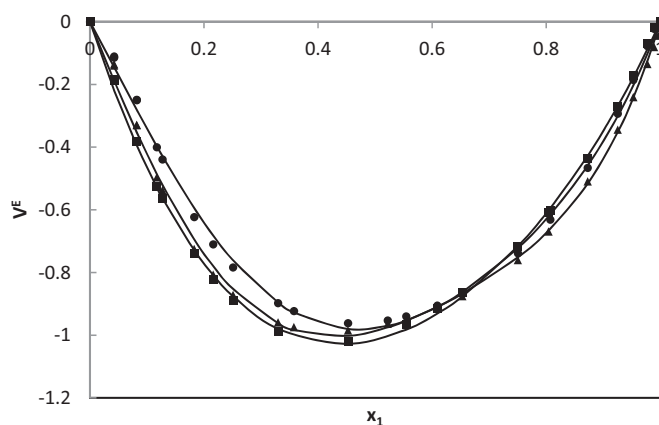


Fig. 7. V^E ($\times 10^{-3} \text{ m}^3 \text{ kmol}^{-1}$) for the binary system methanol + DIPE: ($\blacksquare$) experimental at 298.15 K, ($\bullet$) experimental at 308.15 K; solid curve calculated from Redlich–Kister parameters.

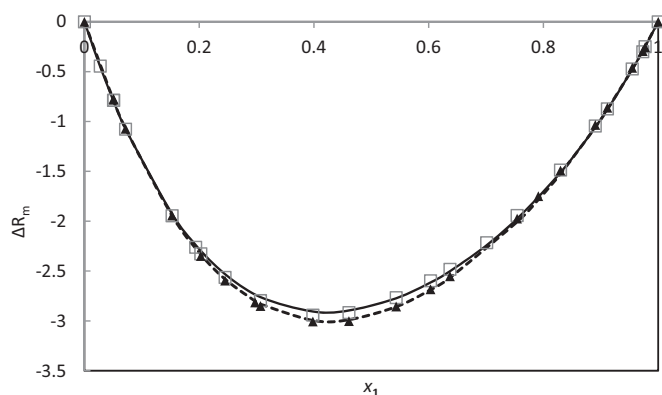


Fig. 6. ΔR ($\times 10^{-3} \text{ m}^3 \text{ kmol}^{-1}$) for the binary system DIPE + IPA. ($\square$) experimental at 293.15 K; ($\blacktriangle$) experimental at 318.15 K; (—) solid curve calculated from Redlich–Kister parameters at 293.15 K and (---) calculated from Redlich–Kister parameters at 318.15 K.

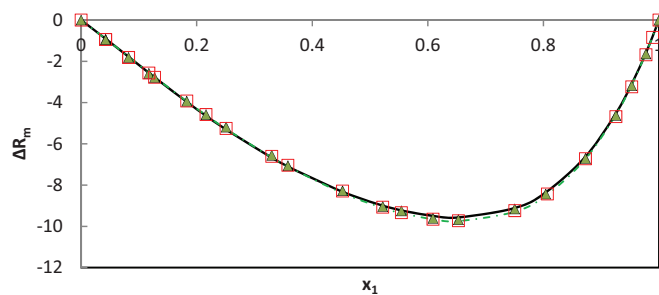


Fig. 8. ΔR ($\times 10^{-3} \text{ m}^3 \text{ kmol}^{-1}$) for the binary system methanol + DIPE. ($\blacktriangle$) experimental at 293.15 K; ($\square$) experimental at 313.15 K; (—) solid curve calculated

Table 14

Fitted parameters of Redlich–Kister equation and standard deviation (SD) for V^E and ΔR for DIPE + IPA system at different temperatures.

T (K)	($\times 10^{-3} \text{ m}^3 \text{ kmol}^{-1}$)	A_1	A_2	A_3	A_4	SD($\times 10^{-3} \text{ m}^3 \text{ kmol}^{-1}$)
293.15	V^E	−1.4525	0.1270	−6.3636	−2.8885	0.0187
	ΔR	11.4056	2.9331	−2.9208	0.1447	0.0225
298.15	V^E	−2.2100	−0.2470	−5.3243	−3.2540	0.0324
	ΔR	−11.6256	3.4932	−3.0119	−1.1170	0.0245
303.15	V^E	−1.2746	0.4146	−6.1044	−3.3824	0.0185
	ΔR	−11.458	3.6471	−2.9087	−1.1370	0.0151
308.15	V^E	−0.4508	1.7263	−6.6500	−3.5330	0.0217
	ΔR	−11.3338	3.9812	−2.8714	−1.1366	0.0114
313.15	V^E	−1.5042	0.373	−6.8651	−3.4924	0.0241
	ΔR	−11.5917	3.3692	−2.7924	−0.3476	0.0157
318.15	V^E	−2.0250	−0.1515	−4.3004	−1.9979	0.0135
	ΔR	−11.7682	3.1030	−2.3698	−0.1124	0.0193

varies with temperature, it does not influence ΔR values significantly.

4. Conclusions

The experimental T – x_1 data for DIPE + IPA and methanol + DIPE systems were measured at atmospheric and sub-atmospheric pressures using a modified Sweitoslawsky ebulliometer. Activity coefficient models, Wilson, NRTL, UNIQUAC, group contribution

method UNIFAC and EoS models PR, SRK with WS mixing rules have been fitted to the experimental data and the optimized model parameters have been reported. Temperature predictions of all the models except UNIFAC are in good agreement with the experimental values, with NRTL model giving the best fit to the experimental data. Density and refractive indices of the pure components and mixtures were measured at different temperatures and the effect of temperature on excess volumes and deviation in molar refractivity is reported.

Table 15Fitted parameters of Redlich–Kister equation and standard deviation (SD) for V^E and ΔR for methanol + DIPE system at different temperatures.

T (K)	$(\times 10^{-3} \text{ m}^3 \text{ kmol}^{-1})$	A_1	A_2	A_3	A_4	SD $(\times 10^{-3} \text{ m}^3 \text{ kmol}^{-1})$
293.15	V^E	−3.9863	−0.0798	−1.0424	0.1547	0.0262
	ΔR	−35.2211	−18.0840	−12.7882	−9.2255	0.0760
298.15	V^E	−4.0588	0.8549	−0.7287	−0.0646	0.0168
	ΔR	−35.5621	−19.4660	−12.5237	−6.8742	0.0620
303.15	V^E	−3.8802	0.6233	−0.9514	0.0406	0.0202
	ΔR	−35.4278	−19.0218	−12.8642	−7.8408	0.0567
308.15	V^E	−3.9453	0.9357	−1.3072	−1.6318	0.0230
	ΔR	−35.4132	−18.5011	−13.8493	−9.6412	0.0586
313.15	V^E	−3.9323	1.0823	−1.3173	−1.6235	0.0380
	ΔR	−35.4099	−18.4692	−13.4130	−9.4410	0.0680
318.15	V^E	−3.9082	0.5036	−0.1669	−1.2279	0.0224
	ΔR	−35.6051	−19.4830	12.3449	−8.0348	0.0541

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